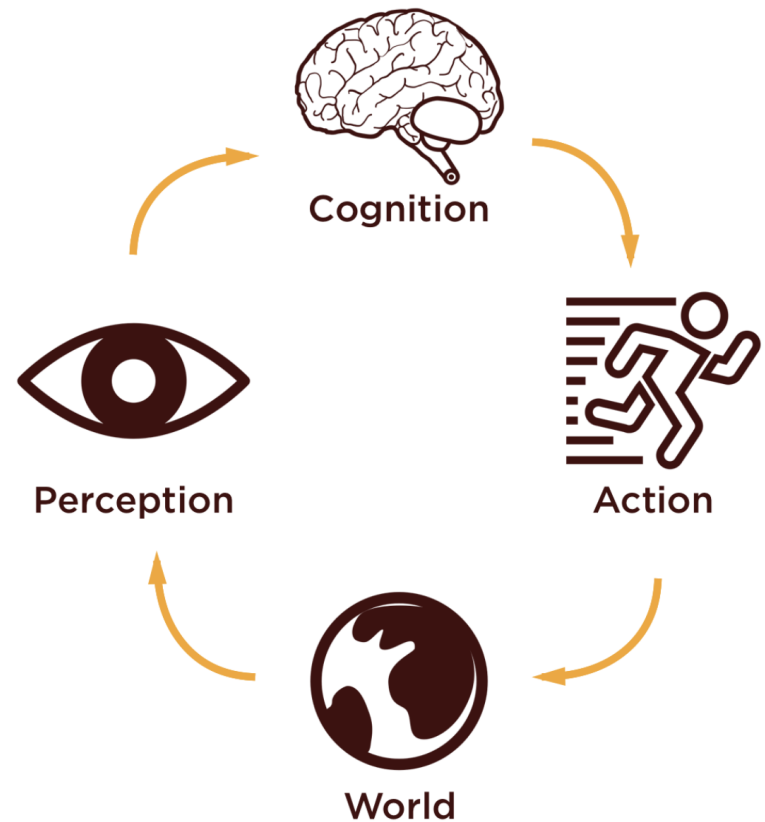


Understanding People: Cognition

CSU44051 Human Factors

Cognition

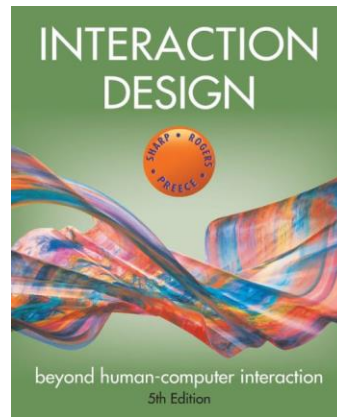
- Interacting with technology is cognitive.
- We need to take into account cognitive processes involved and cognitive limitations of users



How can understanding cognition help HCI?

1. Provides knowledge about what people can and cannot be expected to do
2. Identifies and explains the nature and causes of problems people encounter when using technology
3. Provides theories, modelling tools, guidance, and methods that can lead to the design of better interactive products.

Cognition



Chapter 4: Cognitive Aspects

Cognitive Processes

1. Attention
2. Perception and Recognition 感知和识别
3. Reading, speaking and listening.
4. Memory.
5. Learning.
6. Problem-solving, planning, reasoning, decision making.

Cognitive Processes

1. Attention

2. Perception and Recognition

3. Reading, speaking and listening.

4. Memory.

5. Learning.

6. Problem-solving, planning, reasoning, decision making.

Attention

Selecting things to concentrate on at a point in time from the mass of stimuli around us.

The difficulty of this process for a person depends on:

1. How clear their goal is
2. How salient in their environment the information they need is 在他们的环境中，他们需要的信息有多突出

Focussed and divided attention

- Enables us to be selective in terms of the mass of competing stimuli, but limits our ability to keep track of all events. 使我们能够在大量竞争刺激方面有选择性，但限制了我们跟踪所有事件的能力。

Activity: Find the price of a double room at the Holiday Inn in Bradley

Pennsylvania

Bedford Motel/Hotel: Crinaline Courts

(814) 623-9511 S: \$18 D: \$20

Bedford Motel/Hotel: Holiday Inn

(814) 623-9006 S: \$29 D: \$36

Bedford Motel/Hotel: Midway

(814) 623-8107 S: \$21 D: \$26

Bedford Motel/Hotel: Penn Manor

(814) 623-8177 S: \$19 D: \$25

Bedford Motel/Hotel: Quality Inn

(814) 623-5189 S: \$23 D: \$28

Bedford Motel/Hotel: Terrace

(814) 623-5111 S: \$22 D: \$24

Bradley Motel/Hotel: De Soto

(814) 362-3567 S: \$20 D: \$24

Bradley Motel/Hotel: Holiday House

(814) 362-4511 S: \$22 D: \$25

Bradley Motel/Hotel: Holiday Inn

(814) 362-4501 S: \$32 D: \$40

Breezewood Motel/Hotel: Best Western Plaza

(814) 735-4352 S: \$20 D: \$27

Breezewood Motel/Hotel: Motel 70

(814) 735-4385 S: \$16 D: \$18

Activity: Find the price for a double room at the Quality Inn in Columbia

South Carolina

City	Motel/Hotel	Area code	Phone	Rates	
				Single	Double
Charleston	Best Western	803	747-0961	\$26	\$30
Charleston	Days Inn	803	881-1000	\$18	\$24
Charleston	Holiday Inn N	803	744-1621	\$36	\$46
Charleston	Holiday Inn SW	803	556-7100	\$33	\$47
Charleston	Howard Johnsons	803	524-4148	\$31	\$36
Charleston	Ramada Inn	803	774-8281	\$33	\$40
Charleston	Sheraton Inn	803	744-2401	\$34	\$42
Columbia	Best Western	803	796-9400	\$29	\$34
Columbia	Carolina Inn	803	799-8200	\$42	\$48
Columbia	Days Inn	803	736-0000	\$23	\$27
Columbia	Holiday Inn NW	803	794-9440	\$32	\$39
Columbia	Howard Johnsons	803	772-7200	\$25	\$27
Columbia	Quality Inn	803	772-0270	\$34	\$41
Columbia	Ramada Inn	803	796-2700	\$36	\$44
Columbia	Vagabond Inn	803	796-6240	\$27	\$30

Activity

- Tullis found that the two screens produced quite different results
 - 1st screen - took an average of 5.5 seconds to search
 - 2nd screen - took 3.2 seconds to search
- Why, since both displays have the same density of information (31%)?
- Spacing
 - In the 1st screen the information is bunched up together, making it hard to search
 - In the 2nd screen the characters are grouped into vertical categories of information making it easier

Distraction and interruption

- Users may be distracted or interrupted - “Cup of tea” problem.
- Users should know where they are in the system and what information is being displayed.
- This is particularly important in interfaces based around forms.
- Users should be able to pick up where they left off easily. Security mechanisms like auto-logout can often disrupt this.
- A study with commercial pilots showed that abruptly delivered instructions in descent and approach flight phases resulted in ongoing procedures being 53% more likely to contain errors (Latorella).

Attention

“The *scarce* resource of the 21st century will not be technology; it will be *attention*.”

Mark Weiser

Calm Technology:
Technology should require the
smallest possible amount of
attention.

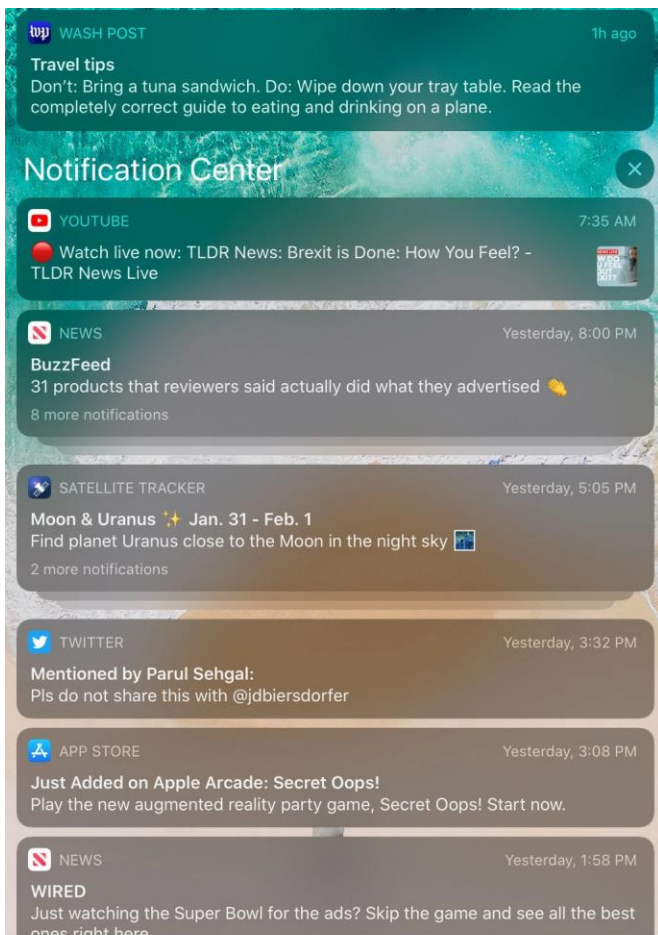


Image: New York Times

Design Implications for Attention

1. Make information salient when it needs attending to at a given stage of a task.
2. Use techniques that make things stand out like colour, ordering, spacing, underlining, sequencing and animation
3. Avoid cluttering the interface with too much information.
4. Consider designing different ways to support effective switching and returning to an interface.

Cognitive Processes

~~1. Attention~~

2. Perception and Recognition

3. Reading, speaking and listening.

4. Memory.

5. Learning.

6. Problem-solving, planning, reasoning, decision making.

Perception and recognition

How information is acquired from the world and transformed into experiences

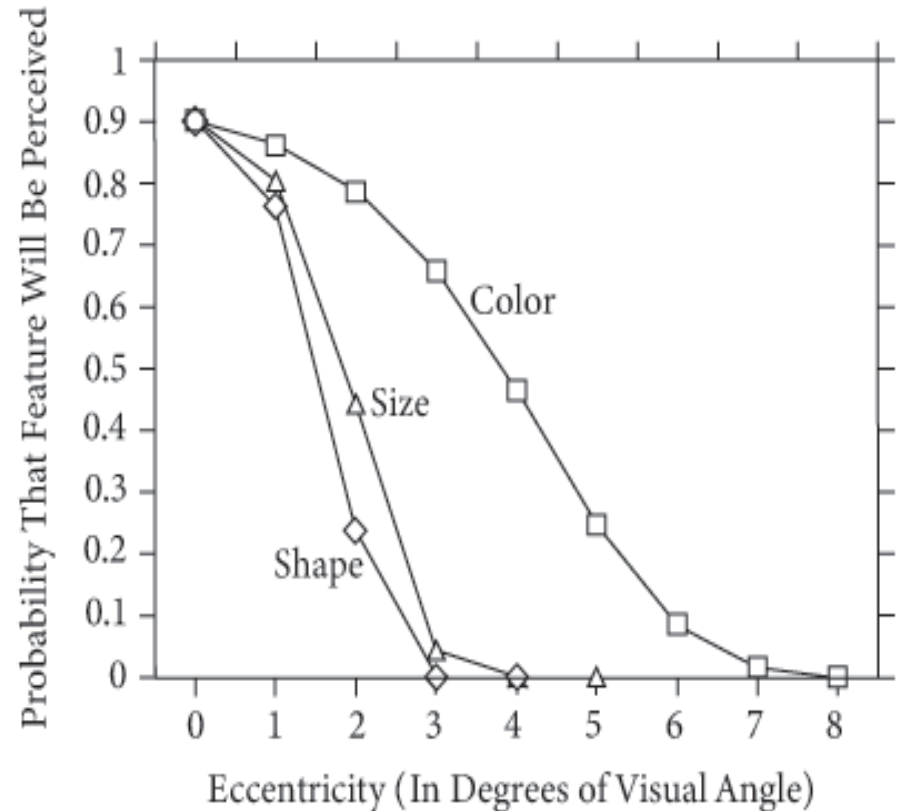
Five senses for information acquisition: vision, hearing, taste, smell, touch

Q. How do we design representations that are readily accessible?

Vision

Foveal vision: Narrow but precise, high-fidelity perception around point of fixation. Spans 2 degrees of angle, roughly the size of your thumb at arm's length.

Peripheral vision: The rest of the field of view, which has eccentrically decreasing fidelity. The farther away from the foveal area, the lower the fidelity.



The farther a feature on display is from the foveal region the less probable it is perceivable

Find Italian

Black Hills Forest
Cheyenne River
Social Science
South San Jose
Badlands Park
Juvenile Justice

Peters Landing
Public Health
San Bernardino
Moreno Valley
Altamonte Springs
Peach Tree City

Jefferson Farms
Psychophysics
Political Science
Game Schedule
South Addition
Cherry Hills Village

Devlin Hall
Positions
Hubard Hall
Fernadino Beach
Council Bluffs
Classical Lit

Results and Stats
Thousand Oaks
Promotions
North Palermo
Credit Union
Wilner Hall

Highland Park
Manchesney Park
Vallecito Mts.
Rock Falls
Freeport
Slaughter Beach

Creative Writing
Lake Havasu City
Engineering Bldg
Sports Studies
Lakewood Village
Rock Island

Sociology
Greek
Wallace Hall
Concert Tickets
Public Radio FM
Children's Museum

Performing Arts
Italian
Coaches
McKees Rocks
Glenwood Springs
Urban Affairs

Rocky Mountains
Latin
Pleasant Hills
Observatory
Public Affairs
Heskett Center

Deerfield Beach
Arlington Hill
Preview Game
Richland Hills
Experts Guide
Neff Hall

Writing Center
Theater Auditions
Delaware City
Scholarships
Hendricksville
Knights Landing

McLeansboro
Experimental Links
Graduation
Emory Lindquist
Clinton Hall
San Luis Obispo

Brunswick
East Millinocket
Women's Studies
Vacant
News Theatre
Candlewood Isle

Grand Wash Cliffs
Indian Well Valley
Online Courses
Lindquist Hall
Fisk Hall
Los Padres Forest

Modern Literature
Studio Arts
Hughes Complex
Cumberland Flats
Central Village
Hoffman Estates

Find French

Webmaster
Russian
Athletics
Go Shockers
Degree Options
Newsletter

Curriculum
Emergency (EMS)
Statistics
Award Documents
Language Center
Future Shockers

Student Life
Accountancy
McKnight Center
Council of Women
Commute
Small Business

Dance
Gerontology
Marketing
College Bylaws
Why Wichita?
Tickets

Geology
Manufacturing
Management
UCATS
Alumni News
Saso

Intercollegiate
Bowling
Wichita Gateway
Transfer Day
Job Openings
Live Radio

Thinker & Movers
Alumni
Foundations
Corbin Center
Jardine Hall
Hugo Wall School

Career Services
Doers & Shockers
Core Values
Grace Wilkie Hall
Strategic Plan
Medical Tech

Educational Map
Physical Plant
Graphic Design
Non Credit Class
Media Relations
Advertising

Beta Alpha Psi
Liberal Arts
Counseling
Biological Science
Duerksen Fine Art
EMT Program

Staff
Aerospace
Choral Dept.
Alberg Hall
French
Spanish

Softball, Men's
McKinley Hall
Email
Dental Hygiene
Tenure
Personnel Policies

English
Graduate Complex
Music Education
Advising Center
Medical School
Levitt Arena

Religion
Art Composition
Physics
Entrepreneurship
Koch Arena
Roster

Parents
Wrestling
Philosophy
Wichita Lyceum
Fairmount Center
Women's Museum

Instrumental
Nursing
Opera
Sports History
Athletic Dept.
Health Plan

Which information structure was better?

Weller (2004) found people took less time to locate items for information that was grouped using a border (2nd screen) compared with using colour contrast (1st screen)

Layout

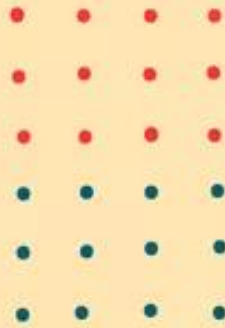
Layout is an important aspect of the design.

- Consistent layout
 - E.g., using different areas of the screen for different purposes can be helpful to the user.
- Grouping items
 - Grouping can be helpful to indicate that items are related in some fashion.
- Whitespace
 - Some argue that too much white space on web pages is detrimental to search process as it makes it hard to find information

Laws of Gestalt

- **Proximity** - items are grouped according to the nearness of their respective parts
- **Similarity** - similar items tend to be grouped
- **Good Continuation** - e.g. straight lines appear to continue as straight lines, curves as curves.
- **Closure/Good Form** - completed items are grouped together
- **Membership character** - a single part of the whole is defined by the context in which it appears.

Examples of the Gestalt Laws



Law of Similarity



Law of Pragnanz or the
Law of Good Figure



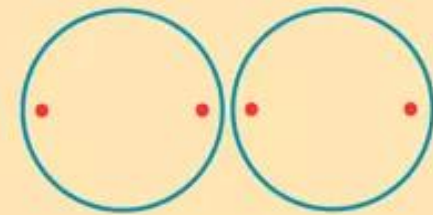
Law of Proximity



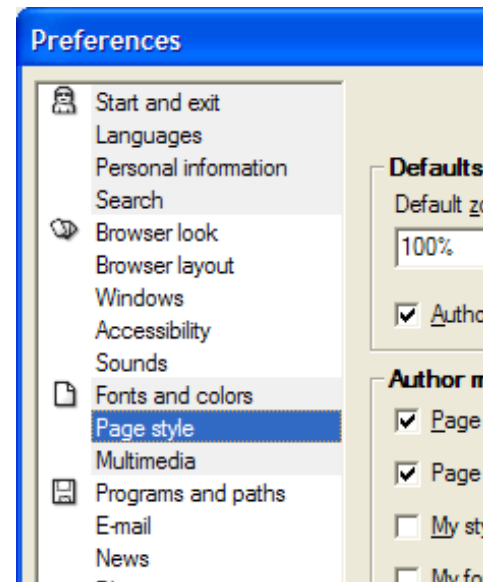
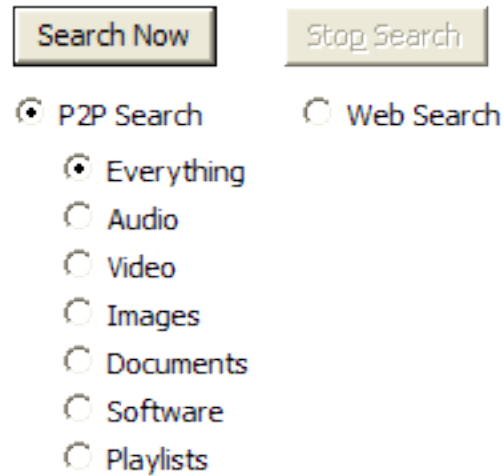
Law of Continuity



Law of Closure



The Law of Common Region



NEWS

Poisoned Navalny discharged from Berlin hospital

The Putin-critic later described his slow recovery in Germany from poisoning with a nerve agent.

1h Europe



- Two hours that saved Navalny's life
- What are Novichok nerve agents?
- Russia's vociferous Putin critic

LIVE
UK hospitality sector labels new rules 'devastating'

5m UK chancellor to update MPs on job protection plans

14m Belgium to reduce quarantine period

23m False claims about testing spreading online



Lukashenko shuts down capital for secret ceremony

The embattled president of Belarus is quietly sworn in for a sixth term despite weeks of protests.

3h Europe



Harry urges US voters to 'reject hate speech'

His "non-partisan" plea has caused controversy, since royals are expected to stay politically neutral.

1h UK



Health chief hails Africa's fight against Covid-19

Africa has had fewer cases than Europe, Asia or the Americas, with numbers continuing to decline.

3h Africa



Assange fiancée 'dreaded making relationship public'

Stella Moris gave birth to the couple's two sons while he was in hiding in the Ecuadorean embassy.

8h World

Over 300 whales die in Australia's worst stranding

4h Australia

Schoolchildren killed in Nigeria tanker explosion

7m Africa

Austrian government sued over Covid at ski resorts

4h Europe

EU plans mandatory migrant pact to 'rebuild trust'

4h Europe

The toughest workplace romance rules in the world?

3h Business

New Zealand eases mask rule as Covid cases drop

1h Asia

Four Seasons' Tommy DeVito dies aged 92

8h US & Canada

Mexico investigating US immigrant 'sterilisations'

3h Latin America & Caribbean

Must See

Guidelines for suppliers:

7. Visual Load

- Ensure that during normal use, the interface should not appear cluttered.
- Consider grouping, alignment, spacing, and gestalt to organise layout for optimum clarity.
- Use as few colours and fonts faces as you need.
- Where possible use the BBC's typeface: BBC Reith.
- Where colour is used to indicate specific meaning, use additional visual treatments to distinguish this further. Do not rely on colour alone to denote difference.
- Use minimal animation (moving or flashing graphical items) and if possible, set as a user preference.
- Ensure there is enough contrast between text and the background. (See the BBC Accessibility and Assistive Technology Standards).

Graphic design example

In 1966, engineers and designers at Dow Chemical working for the National Cancer Institute set out to create an icon for biohazardous materials. They laid out six design criteria. The solution had to be:

1. Striking in form, in order to draw immediate attention
2. Unique and unambiguous to avoid confusion with other symbols
3. Quickly recognizable and easily recalled
4. Easily stencilled
5. Symmetrical, in order to appear identical from all angles
6. Acceptable to groups of varying ethnic backgrounds

Graphic design example

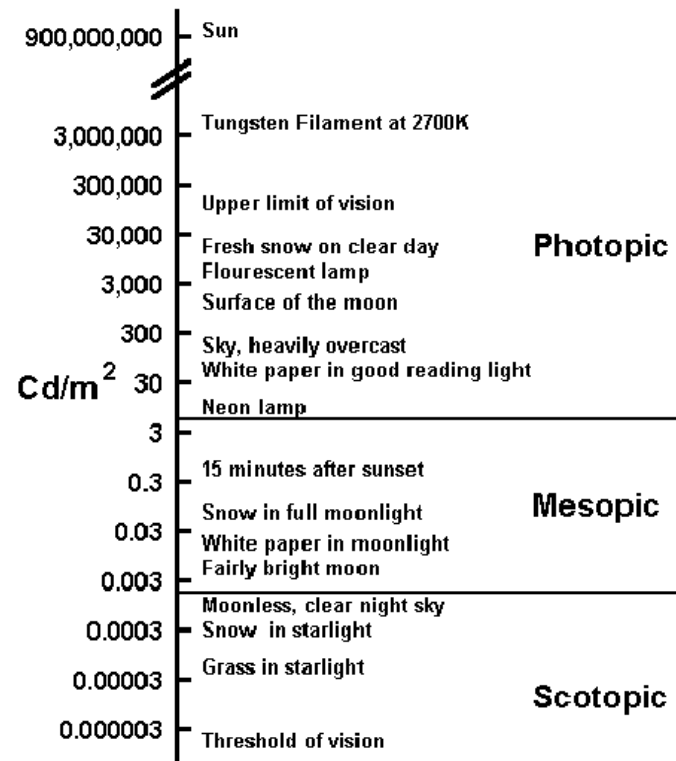
- Showed a set of 24 symbols to 300 people with various amounts of income and formal education from 25 American cities.
- Asked to guess the meaning of each, giving a “meaningfulness score.” A week later, the same participants were shown the original 24 symbols with 36 more and asked to identify which symbols they remembered seeing before.

One scored joint highest in memorability, but the lowest in meaningfulness. So it was unforgettable, but also a had no conflicting interpretation.



Lighting

- 300-500 Lux - office lighting
- 10,000-20,000 Lux - full sunlight, looking towards horizon.



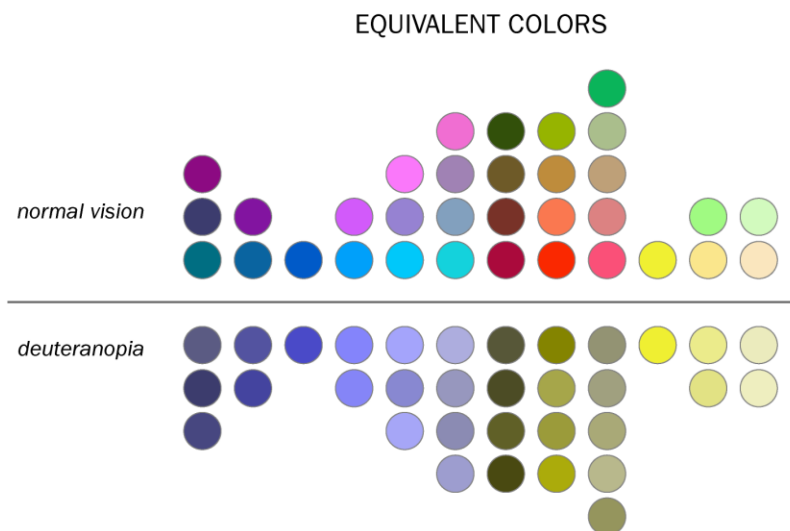
Display readability



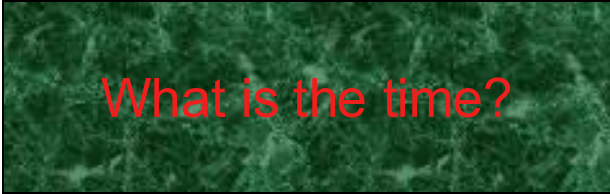
Colour

- Made up of hue, intensity, saturation
- About 8% of men have some form of colour vision deficiency, about 0.5% of women

12-color palette adapted for color blindness



Which is easiest to read and why?



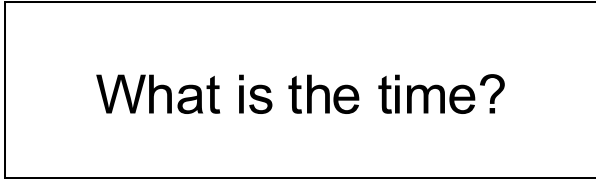
What is the time?



What is the time?



What is the time?

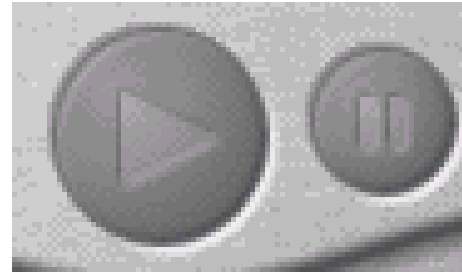


What is the time?



What is the time?

Readability



- Which is better?
- W3C recommends brightness difference and minimum colour difference

Touch

- Provides important feedback about environment.
- May be key sense for someone who is visually impaired.
- Stimulus received via receptors in the skin:
 - thermoreceptors: heat and cold
 - nociceptors: pain
 - mechanoreceptors: pressure (some instant, some continuous)
- Some areas more sensitive than others e.g. fingers.
- Kinesthesia and proprioception - awareness of body position and movement, affects comfort and performance.

Haptic feedback

- Simple haptic feedback may be on/off.
- Responses to keypresses.
- Continuous cycling.
- Pulses.



Surgical simulators



Image: Surgical Science

Haptic screens



Experimental prototypes



Image: Stephen Brewster, CHI 2016 EA.

Movement

- Time taken to respond to stimulus: reaction time + movement time
- Movement time - dependent on age, fitness etc.
- Reaction time - dependent on stimulus type:
 - visual ~ 200ms
 - auditory ~ 150 ms
 - pain ~ 700ms

Movement (cont)

- Fitts' Law (1954) describes the time taken to hit a screen target:

$$M_t = a + b \log_2(D/S + 1)$$

- Where ***a*** and ***b*** are empirically determined constants
 - M_t is movement time
 - D is Distance
 - S is Size of target

⇒ targets as large as possible, distances as small as possible

Design Implications: Perception and Recognition

1. Icons and other graphical representations should enable users to readily *distinguish* their meaning
2. Bordering and spacing are effective visual ways of grouping information
3. Sounds should be audible and distinguishable
4. Text should be legible and distinguishable from the background
5. Haptic feedback should be used judiciously

Inclusive design

Support Card | The Persona Spectrum

We use a Persona Spectrum to understand related limitations across a spectrum of permanent, temporary, and situational disabilities. It's a quick tool to help foster empathy and to show how a solution can scale to a broader audience.

	Permanent	Temporary	Situational
Touch	 One arm	 Arm injury	 New parent
See	 Blind	 Cataract	 Distracted driver
Hear	 Deaf	 Ear infection	 Bartender
Speak	 Non-verbal	 Laryngitis	 Heavy accent

Inclusive design -
accommodate the widest
possible group of people.

Can make the design better
for everyone.

Cognitive Processes

~~1. Attention~~

~~2. Perception and Recognition~~

3. Reading, speaking and listening.

4. Memory.

5. Learning.

6. Problem-solving, planning, reasoning, decision making.

Reading, Speaking, and Listening

The ease with which people can read, listen, or speak differs:

- Many prefer listening to reading
- Reading can be quicker than speaking or listening
- Listening requires less cognitive effort than reading or speaking
- People with Dyslexia have difficulties understanding and recognising written words

Reading

- Several stages:
 - visual pattern perceived
 - decoded using internal representation of language
 - interpreted using knowledge of syntax, semantics, pragmatics
- Reading involves saccades and fixations
- Perception occurs during fixations
- Word shape is important to recognition
- Negative contrast (dark on light) improves reading from computer screen

Audio in Interfaces

- How might we make use of sound in an interface?
 - Getting the user's attention
 - Continuous status information
 - Confirmation
 - Hands-free eyes-free interaction



Design Implications: Reading, Speaking, and Listening

1. Speech-based menus and instructions should be short
2. Accentuate the intonation of artificially generated speech voices
 - They are harder to understand than human voice
3. Provide opportunities for making text large on a screen

Cognitive Processes

~~1. Attention~~

~~2. Perception and Recognition~~

~~3. Reading, speaking and listening.~~

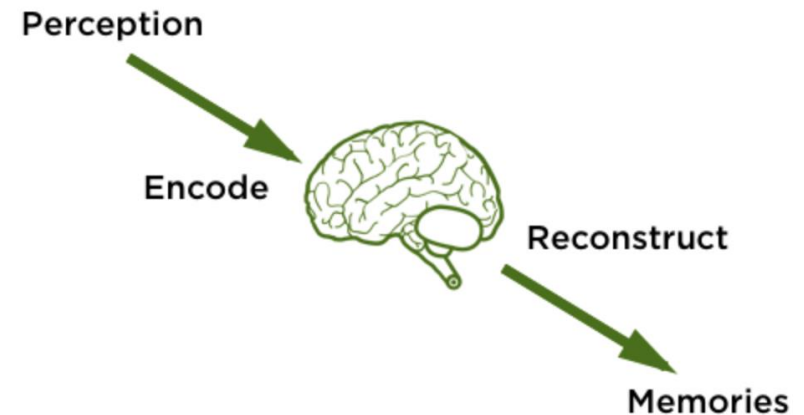
4. Memory.

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Memory

- Involves recalling various kinds of knowledge that allow people to act appropriately
 - For example, recognizing someone's face or remembering someone's name
- First encode and then retrieve knowledge
- We don't remember everything - involves filtering and processing what is attended to



Processing in memory

- Encoding is first stage of memory
 - determined by which information is attended to in the environment and how it is interpreted
- The more attention paid to something, and the more it is processed in terms of thinking about it and comparing it with other knowledge...

The more likely it is to be remembered

- *For example, when learning about HCI, it is much better to reflect upon it, carry out exercises, have discussions with others about it, and write notes than just passively read a book or listen to a lecture.*

Context is important

- Context affects the extent to which information can be subsequently retrieved
- Sometimes it can be difficult for people to recall information that was encoded in a different context
 - e.g., You are on a train and someone comes up to you and says hello. You don't recognise him for a few moments but then realise it is one of your neighbours. You are only used to seeing your neighbour near where you live and seeing him out of context makes him difficult to recognise initially

Memory

There are three types of memory function:

1. Sensory memories (buffers for stimuli: visual : iconic, auditory : echoic, touch : haptic)
2. Short-term (STM)
3. Long-term (LTM)

Short-term memory (STM)

- Scratch-pad for temporary recall
 - rapid access ~ 70ms
 - rapid decay ~ 200ms
 - limited capacity - 7 ± 2 chunks

The problem with the classic '7 $\pm$ 2'

- George Miller's theory of how much information people can remember
- People's immediate memory capacity is very limited
- Many designers have been led to believe that this is useful finding for interaction design

What some designers get up to...

- Present only 7 options on a menu
 - Display only 7 icons on a tool bar
 - Have no more than 7 bullets in a list
 - Place only 7 items on a pull down menu
 - Place only 7 tabs on the top of a website page
- But this is wrong? Why?



Why?

- Inappropriate application of the theory
- People can scan lists of bullets, tabs, menu items till they see the one they want
- They don't have to recall them from memory having only briefly heard or seen them
- Sometimes a small number of items is good design.
 - E.g. phone or smartwatch displays.
- Depends on task and available screen estate

Long-term memory (LTM)

- Repository for all our knowledge
 - slow access ~ 1/10 second
 - slow decay, if any
 - huge or unlimited capacity
- Two types
 - episodic - serial memory of events
 - semantic - structured memory of facts, concepts, skills
 - information in semantic LTM derived from episodic LTM.

LTM - Storage of information

- Rehearsal
 - information moves from STM to LTM
- Total time hypothesis
 - amount retained proportional to rehearsal time
- Distribution of practice effect
 - optimised by spreading learning over time
- Structure, meaning and familiarity
 - information easier to remember

LTM - Forgetting

- Decay
 - information is lost gradually but very slowly
- Interference
 - new information replaces old: retroactive interference (eg. phone number)
 - old may interfere with new: proactive inhibition
- May not forget at all: memory is selective...
- Affected by emotion - can subconsciously “choose” to forget
- Impacted by medical conditions.

LTM - retrieval

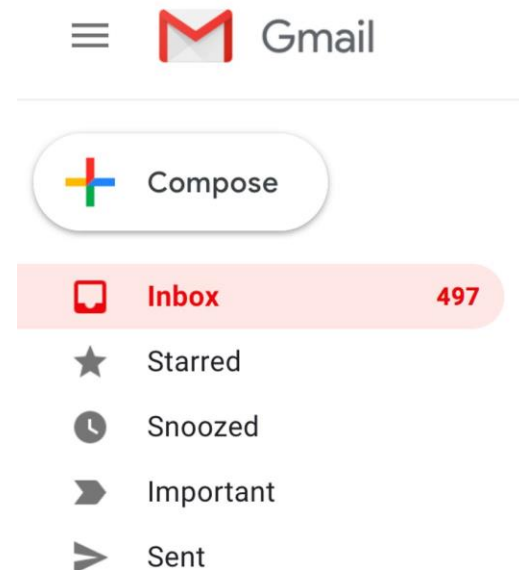
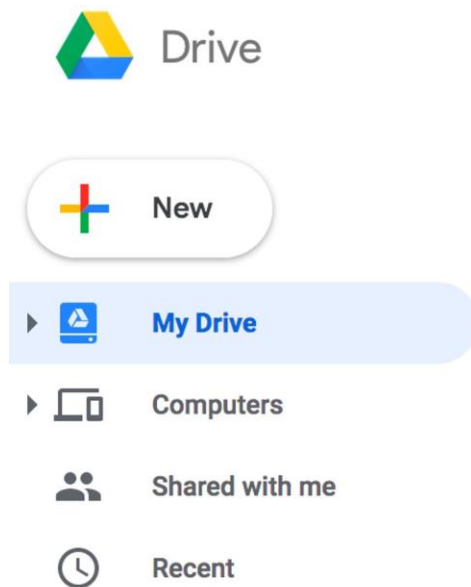
- Recall
 - information reproduced from memory
 - can be assisted by cues, e.g. categories, imagery
- Recognition
 - information gives knowledge that it has been seen before
 - less complex than recall - information is cue

Recognition versus recall

- Command-based interfaces require users to recall from memory a name from a possible set of hundreds
- GUIs provide visually-based options that users need only browse through until they recognise one
- Web browsers, music players, etc., provide lists of visited URLs, song titles etc., that support recognition memory

Reducing memory load - consistency

- Consistency is useful as it can allow the user to predict what will happen when they carry out a certain action.
- Likewise they can have expectations on where certain information can be found.
- This can work across a given application, suite of applications, or an entire technology platform.





Personal information management

- Personal information management (PIM) is a growing problem for most users
 - Accumulate vast numbers of documents, images, music files, video clips, emails, attachments, bookmarks, etc.,
 - Where and how to save them all, then remembering what they were called and where to find them again
 - Naming most common means of encoding them
 - But can be difficult to remember, especially when you have 10,000s
 - How might such a process be facilitated taking into account people's memory abilities?

Personal information management

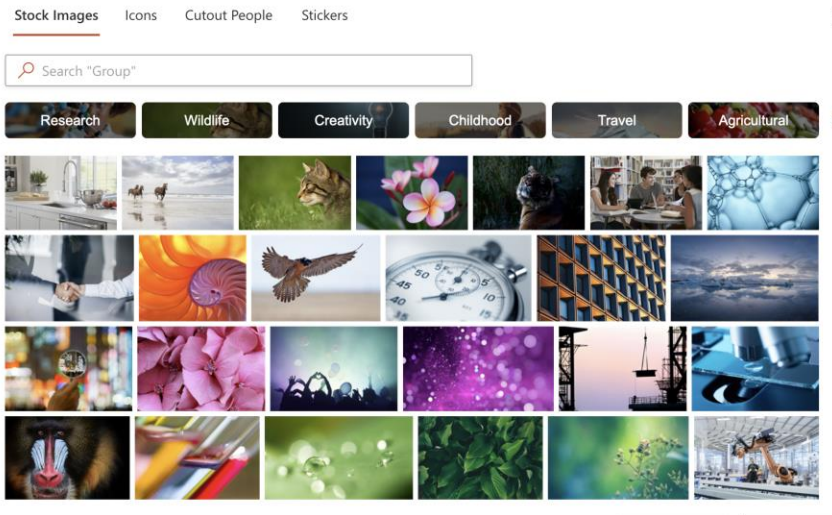
- Three interdependent processes model (Bergman and Whittaker 2016) to help people manage their stuff :
 1. How to decide what stuff to keep
 2. How to organize it when storing
 3. Which strategies to use to retrieve it later
- Retrieval involves 2 processes
 - recall-directed and recognition-based scanning
- Strong preference for scanning.



Figure 3.3 Apple's Spotlight search tool

Personal information management

- Optimise both kinds of memory processes
 - e.g., Search box and history list
- Help users encode files in richer ways
 - Provide them with ways of saving files using colour, flagging, image, flexible text, time stamping, etc



Memory load

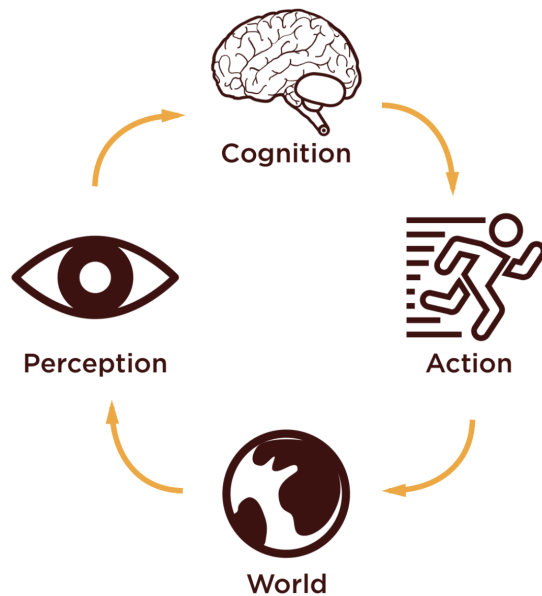
- Services like online/mobile and phone banking now require users to provide multiple pieces of information or devices to access their account
 - For instance, postcode, birthplace, a memorable date, first school attended
 - Known as multifactor authentication (MFA)
- Why?
 - Increased security concerns
- Password managers, such as LastPass, have been developed that require only one master password
 - reduces stress and memory load on users
- Passwords could become less common with the widespread use of biometrics and computer vision algorithms

Digital Forgetting

- When might you wish to forget something that is online?
 - When you break up with a partner
 - Emotionally painful to be reminded of them through shared photos, social media, and so on.
- Sas and Whittaker (2013) suggest ways of harvesting and deleting digital content
 - For example, making photos of ex into an abstract collage
 - Helps with closure
- ⑩ Researchers are also looking at the role of digital technologies in bereavement and grief.

Cognition

- Interacting with technology is cognitive.
- We need to take into account cognitive processes involved and cognitive limitations of users



- ~~1. Attention~~
- ~~2. Perception and Recognition~~
- ~~3. Reading, speaking and listening.~~
- ~~4. Memory~~
5. Learning
6. Problem-solving, planning, reasoning, decision making.

Design implications: Memory

1. Reduce cognitive load by avoiding long and complicated procedures for carrying out tasks.
2. Avoid overloading short term memory (e.g. speech menus).
3. Design interfaces that promote recognition rather than recall.
4. Provide users with various ways of labelling digital information to help them easily identify it again.
 - For example, folders, categories, colour, flagging, and time stamping

Cognitive Processes

~~1. Attention~~

~~2. Perception and Recognition~~

~~3. Reading, speaking and listening.~~

~~4. Memory~~

5. Learning

6. Problem-solving, planning, reasoning, decision making.

Learning

- Involves the accumulation of skills and knowledge involving memory
- Two main types:
 - Incidental learning (for example, recognizing people's faces, what you did today)
 - Intentional learning (for instance, studying for an exam, learning to cook)
 - Intentional learning is much harder!
 - Many technologies have been developed to help (for example, multimedia, animations, VR)
- Many people find it hard to learn by following instructions in a manual
- Many people prefer to learn by doing

Design implications

1. Design interfaces that encourage exploration
 - Users that explore interfaces end up more efficient at task performance.
2. Design interfaces that constrain and guide learners
 - Think about frequency of use.
3. Dynamically linking concepts and representations can facilitate the learning of complex material

Cognitive Processes

~~1. Attention~~

~~2. Perception and Recognition~~

~~3. Reading, speaking and listening~~

~~4. Memory~~

~~5. Learning~~

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6. Problem-solving, planning, reasoning, decision making.

Problem-solving, planning, reasoning, and decision-making

- All these processes involve *reflective* cognition
 - For example, thinking about what to do, what the options are, and the consequences
- Often involves conscious processes, discussion with others (or oneself), and the use of artifacts
 - Such as maps, books, pen and paper
- May involve working through different scenarios and deciding which is best option.
- Make "just good enough" decisions based on "fast and frugal" heuristics, or risk averse, relying on assistance to make decisions?
- Weighing up alternatives

Design implications

1. Provide information and help pages that are easy to access for people who wish to understand more about how to carry out an activity more effectively.
2. Use simple and memorable functions to support rapid decision-making and planning

Cognitive Processes

1. Attention

2. Perception and Recognition

3. Reading, speaking and listening.

4. Memory

5. Learning

6. Problem-solving, planning, reasoning, decision making.